

Online House Rent App

Technical Architecture

Frontend: React.js, Material UI, Bootstrap, Axios

Backend: Node.js, Express.js, Socket.io, REST API

Database: MongoDB

System Features & Implementation Details

User Registration & Profile Creation

- Secure sign-up using email and password authentication.
- Stores user details such as name, contact information, profile image, and user type (Tenant / Landlord / Agent).
- User profiles help manage property listings and booking activities.

Property Browsing & Search Filtering

- Users can search and filter properties by location, price range, property type, and availability status.
- Real-time property updates provide users with the latest listing information.

Property Listing & Management

- Landlords can add property details such as title, description, address, rent amount, amenities, and images.

- Property availability can be updated dynamically.
- Supports image uploads and detailed property descriptions for better user experience.

Booking & Reservation System

- Users can view available properties and submit booking requests.
- Booking information includes booking dates, status updates, and tenant details.
- Automated notifications inform users regarding booking confirmations, cancellations, and updates.

Landlord Dashboard

- Property owners can view and manage listed properties.
- Allows monitoring of bookings, tenant requests, and property status updates.
- Provides secure management of rental information.

Admin Controls & Verification

- Admin verifies users and property listings to ensure authenticity.
- Provides monitoring and management tools for properties and users.
- Generates reports and analytics for system administration.

User Flow Documentation

1. Tenant / Ordinary User

Role: Search properties, book rentals, communicate with landlords, and manage profile information.

Registration and Login:

Create an account using email and password and log in securely.

Property Search:

Browse properties using filters such as location, price, and property type.

Booking Request:

Select a property, view detailed information, and submit a booking request.

Booking Tracking:

Track booking status and receive real-time notifications regarding approval or updates.

Profile Management:

Manage personal information and booking history.

2. Landlord / Property Owner

Role: Add properties, manage bookings, and communicate with tenants.

Registration and Login:

Register and log in using email and password.

Property Management:

Add, edit, update, or remove property listings and upload images.

Booking Management:

View tenant booking requests and approve or reject reservations.

Tenant Interaction:

Communicate with tenants regarding property details and rental agreements.

3. Admin

Role: Manage the overall platform, monitor users, and maintain system security.

Management and Monitoring:

Monitor all activities related to users, properties, and bookings.

Property Verification:

Verify listed properties and remove invalid or fraudulent information.

User and Landlord Management:

Manage account details and enforce platform policies and privacy regulations.

System Improvement:

Implement security enhancements and improve platform functionality based on user feedback.

MVC Architectural Pattern – House Rent App

The backend application follows the **Model-View-Controller (MVC)** architectural pattern to improve maintainability, scalability, and modular development.

Layers

Model Layer (Data Layer)

Handles database-related operations using Mongoose schemas in MongoDB.

Examples:

- User Model
- Property Model
- Booking Model
- Booking History Model

Controller Layer

Acts as an intermediary between requests and database operations.

Responsibilities:

- Handle user registration/login

- Manage property listings
- Process booking requests
- Update booking status

View Layer (Routing Layer)

Implemented using REST API routes that define application endpoints.

Examples:

- GET → Retrieve property data
- POST → Add properties or create bookings
- PUT → Update property and booking information
- DELETE → Remove properties or cancel bookings

Advantages

- **Separation of Concerns:** Improves code readability and organization.
- **Scalability:** New modules and features can be added easily.
- **Reusability:** Business logic can be reused across components.
- **Testing:** Each layer can be tested independently.
- **Collaboration-Friendly:** Developers can work simultaneously on different modules.

Roles and Responsibilities

User Registration:

Users register with email and password to access rental services.

Property Browsing:

Users browse and search available properties with filtering options.

Property Listing:

Landlords add and manage property details with images and rental information.

Booking Management:

Users submit booking requests and landlords process them.

Admin Monitoring:

Admins verify users and properties, monitor activities, and maintain system security.